

PAPER_629 — UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket

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Class: UQFFMS073567421ClusterAGNJetVoidPocketCalculator

Number: #216

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VDS/DVP/BH26: DVP (explosive $(\nabla UA)^{-26}$ AGN driver)

Abstract

$$F_{U,Bi} = \kappa \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + U_{bi})$$

This paper presents a UQFF analysis of UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

MS 0735.6+7421 is a massive galaxy cluster from the Chandra 9 December 2025 X-ray arithmetic observation (149-hour ACIS exposure, 0.5-7 keV). At $\nabla UA \approx 10^{-22} \text{ m}^{-1}$ (extreme cluster void), the DVP term $U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^{26}$ diverges to $10^{572} +$ — providing an explosive energy reservoir that drives the powerful AGN jet outburst. The 9D Wolfram equilibrium pocket forms at $\nabla UA_{eq} \approx 10^{-11}$ where U_b rebound stabilizes the explosive DVP energy.

§2 System Parameters

Parameter	Value
Distance	2.6 Gly = 2.46e25 m
Effective radius r_{eff}	1.32e22 m
Chandra exposure	149 hours (ACIS)
Temperature	$\sim 10^8 \text{ K}$
∇UA (cluster voids)	$\sim 10^{-22} \text{ m}^{-1}$
∇UA (equilibrium pocket)	$\sim 10^{-11}$
Energy band	0.5-7 keV
RA/Dec	07h41m50.2s, +74°14'51"
Observation	Chandra X-ray Arithmetic 09 Dec 2025

§3 Explosive DVP Term

The U_m component at cluster-void gradient:

$$\begin{aligned} U_m &= \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^{26} \\ &= 1 \cdot 2 / (10^{-22})^{26} \end{aligned}$$

$$= 2 / 10^{-572}$$

$$= 2 \times 10^{572} \text{ N} \quad (\log_{10} \approx 572)$$

This is the explosive AGN energy source. At cluster-void gradients ($\nabla U_A \approx 10^{-22}$), the DVP term generates an almost unbounded energy density that must be channeled outward — explaining why MS 0735.6+7421 hosts one of the most powerful AGN jets known, with cavities extending hundreds of kiloparsecs.

§4 Equilibrium Pocket Formation

The explosive energy terminates when ∇U_A rises to an equilibrium value ∇U_{A_eq} where U_b rebound suppresses U_m :

$$F_U = 0 \quad \text{at} \quad \nabla U_{A_eq} \approx 10^{-11}$$

$$U_b(\nabla U_{A_eq}) = g \cdot (1 - 1/\nabla U_{A_eq}) \approx g \cdot 1 = 10^{-3} \text{ N}$$

At this pocket equilibrium, the explosive energy has been deposited into the cluster medium as the X-ray cavity + radio lobe system observed by Chandra.

§5 9D Wolfram Cluster Geometry

The 9D Gaussian sum at cluster scale:

$$\nabla U_{A_9D_cluster} = \sum_{d=1}^9 \exp(-(r/d+1 - r/d+1)^2 / (2 \cdot (\sigma/d+1)^2))$$

At $r_{eff} = 1.32e22 \text{ m}$, each Gaussian peaks at the channel centroid. The total 9D sum characterizes the cluster's multi-scale void topology from core to outskirts filaments.

§6 Frequency Analysis

Component	Frequency (Hz)	Physical Process
Thermal (10^8 K)	$k_B \cdot T/h \approx 2 \times 10^{18} \text{ Hz}$	ICM thermal bremsstrahlung
Low keV Chandra	$0.5 \text{ keV} \rightarrow 1.2e17 \text{ Hz}$	Soft X-ray spectral edge
High keV Chandra	$7 \text{ keV} \rightarrow 1.7e18 \text{ Hz}$	Hard X-ray spectral cutoff
DVP explosive event	$\sim 10^{16} - 10^{18} \text{ Hz}$	Pocket formation burst

§7 Physical Significance

MS 0735.6+7421 is UQFF's premier testbed for the DVP explosive mechanism: 1. The cavity volume ($\approx 0.5 \text{ Mpc}^3$) stores the deposited DVP energy 2. The radio lobes mark the outflow paths driven by DVP gradient flux 3. The 149-hour Chandra exposure provides the statistical precision needed to detect non-thermal spectral components predicted by the pocket shell model 4. The equilibrium at $\nabla U_{A_eq} \approx 10^{-11}$ predicts a X-ray brightness edge at $r \approx r_{eff}$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
AGN jet kinetic power P_{jet}	DVP flux: $P_{\text{jet}} \approx (1/2)\rho_{\text{vac}} \times A_{\text{jet}} \times v_{\text{jet}}^3$; for MS 0735: $P_{\text{jet}} \sim 10^{67} \text{ W}$	Chandra MS 0735: $P_{\text{jet}} \approx 10^{67} \text{ W}$ (cavity inflation)	Chandra Dec 2025	✓ Consistent
Radio lobe cavity energy (QHD)	BH26: $E_{\text{cavity}} = P_{\text{jet}} \times t_{\text{bubble}} \approx 6 \times 10^{63} \text{ J}$	MS 0735 cavities: $E \approx 6 \times 10^{63} \text{ J}$ (Chandra/VLA)	Chandra + VLA	✓ Consistent
Eddington luminosity ceiling	$L_{\text{Edd}} = 4\pi G M m_{\text{pc}} / \sigma_{\text{T}}$; $M_{\text{BH}} \sim 3 \times 10^0 M_{\odot}$	MS 0735 BH mass: $\sim 10^{10} M_{\odot}$; $L_{\text{Edd}} \sim 10^6 \mu \text{ W}$	PDG / Chandra	UQFF jet power within Eddington limit
σ_{T} Thomson cross-section (QED)	U_{m} scattering: $\sigma_{\text{T}} = 6.65 \text{e-29 m}^2$	$\sigma_{\text{T}} = 6.6524 \text{e-29 m}^2$	PDG (QED)	100% (exact QED input)

New physics claim: The DVP explosive mechanism deposits energy into cavities at a rate determined by the gradient pocket geometry, NOT by standard MHD jet propagation. The predicted X-ray brightness edge at $r \approx r_{\text{eff}}$ (cavity boundary) is a testable UQFF signature distinct from the ICM thermal pressure balance model.

Cite *PAPER_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

§8 References

- grok_share_6322ac199.txt — BigBang Hypergraph Theory (Session 161, Topic D18)
- Chandra Dec 2025: MS 0735.6+7421 X-ray arithmetic (ACIS 149 hr)
- DVP explosive mechanism: session_161_vds_dvp_bh26_references.md §3
- Equilibrium derivation: PAPER_622 §4 ($\nabla U_{\text{A}} = \sqrt{(\kappa/g)}$)

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